

BookMyShow — Core API Specifications

1. Authentication & Authorization Flow

- Token Verification:** Every client request to protected endpoints must supply an `Authorization: Bearer <JWT>` header.
- Gateway Enforcement:** The API Gateway intercepts incoming requests and validates the JWT signature and expiration against the Auth Provider's public key (cached locally via JWKS).
- Context Propagation:** Upon successful validation, the Gateway extracts claims (such as `userId` and `role`) and injects trusted internal headers (e.g., `X-User-Id`) before routing to downstream microservices.
- Access Control & Errors:** Unauthenticated requests are rejected immediately at the Gateway with `401 Unauthorized`. If an authenticated user attempts an operation on a resource they do not own (e.g., cancelling another user's booking), the service rejects the request with `403 Forbidden`.

2. Core REST APIs

GET `/movies?city={city}&date={date}&language={language}`

Fetches movies, theaters, and available shows matching city, date, and optional language filters.

QUERY PARAMETERS

Parameter	Type	Required	Description
<code>city</code>	string	Yes	City name or identifier (e.g., <code>delhi</code>)
<code>date</code>	string	Yes	Target date in <code>YYYY-MM-DD</code> format
<code>language</code>	string	No	Language filter (e.g., <code>english</code> , <code>hindi</code>)

RESPONSE SCHEMA (`200 OK`)

```
{
  "city": "delhi",
  "date": "2026-10-01",
  "movies": [
    {
      "movieId": "mov_101",
      "title": "Inception",
      "language": "English",
      "durationMinutes": 148,
      "theaters": [
        {
          "theaterId": "th_201",
          "theaterName": "PVR Rivoli",
          "shows": [
            {
              "showId": "sh_301",
              "screenName": "Audi 1",
              "startTime": "18:30",
              "availableSeats": 42,
              "ticketPrice": 350.00
            }
          ]
        }
      ]
    }
  ]
}
```

STATUS CODES & ERROR HANDLING

- `200 OK` : Successful query. Returns matching movies and shows (or empty array if none found).
- `400 Bad Request` : Missing mandatory parameters (`city` or `date`) or invalid date format.
- `500 Internal Server Error` : Internal search service failure.

IDEMPOTENCY

`GET` operations are safe and idempotent by HTTP specification. Repeating the request does not alter server state.

POST /bookings

Reserves selected seats for a show, processes payment, and confirms the booking.

REQUEST HEADERS

- Authorization: Bearer <token>
- Idempotency-Key: <UUID>

REQUEST BODY

```
{
  "showId": "sh_301",
  "userId": "usr_501",
  "seats": ["A1", "A2"],
  "paymentMethod": "UPI"
}
```

RESPONSE SCHEMA (201 CREATED)

```
{
  "bookingId": "bkg_901",
  "showId": "sh_301",
  "userId": "usr_501",
  "seats": ["A1", "A2"],
  "totalAmount": 700.00,
  "status": "CONFIRMED",
  "bookingTime": "2026-10-01T18:35:00Z"
}
```

STATUS CODES & ERROR HANDLING

- 201 Created : Seats successfully reserved, payment processed, and booking confirmed.
- 400 Bad Request : Malformed JSON payload or missing mandatory fields.
- 401 Unauthorized : Missing or invalid authentication token.
- 409 Conflict : One or more selected seats are already booked or held by another user.
- 422 Unprocessable Entity : Payment declined or failed via payment gateway.
- 500 Internal Server Error : Database error or server failure.

IDEMPOTENCY REQUIREMENTS

The client generates a unique UUID in the Idempotency-Key header. The BookingService caches the key and response in Redis with a 24-hour expiration. If a client retries due to a network timeout, the server returns the cached response rather than creating a duplicate booking or initiating a second charge.

DELETE /bookings/{bookingId}

Cancels an existing booking, releases reserved seats back to the available pool, and initiates a refund.

PATH PARAMETERS

Parameter	Type	Required	Description
bookingId	string	Yes	Unique identifier of the booking to cancel

RESPONSE SCHEMA (200 OK)

```
{
  "bookingId": "bkg_901",
  "status": "CANCELLED",
  "refundAmount": 700.00,
  "message": "Booking cancelled successfully. Refund initiated."
}
```

STATUS CODES & ERROR HANDLING

- 200 OK : Booking successfully cancelled and refund initiated.
- 400 Bad Request : Show has already started or past the permissible cancellation window.
- 401 Unauthorized : Missing or invalid authentication token.
- 403 Forbidden : Authenticated user does not have permission to cancel this booking.
- 404 Not Found : Booking ID does not exist.
- 500 Internal Server Error : Server failure during seat release or refund initiation.

IDEMPOTENCY REQUIREMENTS

DELETE is idempotent. Repeated calls with the same bookingId will consistently return the cancelled state without issuing duplicate refunds or corrupting seat states.